

Towards Corrected Estimators of Epistemic Uncertainty Measures

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Abstract

The epistemic uncertainty associated with the predictions of an ensemble of machine learning models is most often quantified by a measure of expected divergence between the ensemble and a new model drawn from the model posterior. In practice, the expectation over the new model is often intractable and approximated by an average over the models in the ensemble. In this paper, we examine the bias of this approximation. We find that bias-corrected uncertainty estimators yield reductions of up to 21% in mean squared estimation error on Out-of-Distribution samples and 13% on In-Distribution samples on computer vision tasks. However, we also find that corrected estimators don't significantly outperform uncorrected estimators on uncertainty quantification-related tasks, whereas the estimation target does outperform uncorrected estimators. Hence, corrected epistemic uncertainty estimators should be considered by practitioners whenever the magnitude of epistemic uncertainties is meaningful, such as when comparing aleatoric and epistemic uncertainty, or when estimating the expected loss of a prediction. In order to improve downstream task performance, accurate estimation of the difference in uncertainty between different test instances will require further research.

1 Introduction

The quantification of the uncertainty associated with predictions made by machine learning models has gained a lot of traction recently. In particular, the appropriateness of the additive decomposition of predictive uncertainty into its aleatoric and epistemic components has been debated [23, 10] and modifications, as well as generalizations, have been proposed [7, 11]. Aleatoric uncertainty (AU) describes the uncertainty present in a model's predictions due to the inherent noise present in the data-generating process. Epistemic uncertainty (EU) on the other hand, refers to the uncertainty about the true data-generating process, and can therefore be reduced by collecting more training data [20]. The definition of EU therefore arises naturally from a probability distribution over model parameters, i.e. a distribution over distributions which describe the outcome (often called second-order distribution) [9].

Recently, a scoring-rule based decomposition of uncertainty has been proposed [7, 11]. In a special case, this approach allows us to recover the classical entropy-based uncertainty decomposition [2], and can therefore be seen as a generalization thereof. It has also been empirically demonstrated that different scoring rules excel at different downstream tasks [17, 8], opening up yet another new line of investigation in the field. The definition of scoring-rule based uncertainty however, as usual in uncertainty quantification, does generally not allow us to compute the uncertainty exactly for complex models, as the involved expectation over the model posterior tends to be intractable, e.g. for deep ensembles [14]. Before research on uncertainty quantification proceeds with the empirical comparison of various scoring rules, we thus deem it important to revisit the statistical estimation of scoring rule-based uncertainty.

Current practice uses resubstitution estimators, which replace all expectations over second-order distributions by sample averages using the same members of the same ensemble. This approach is naturally associated with an underestimation of the true epistemic uncertainty. While the standard estimate of AU is unbiased, its relative contribution to the total uncertainty (TU) will naturally appear greater due to the underestimated EU.

In this paper, we make advancements on EU estimation through following contributions:

- We clarify the estimation target and derive bias corrections for log-loss and Brier score uncertainty.
- We study the effect of bias correction on estimation error and downstream task performance.

2 Background & Related Work

We primarily consider the classification setting, though recent work has shown how to extend the usage of scoring rules for uncertainty quantification to regression [3]. In the classification case, we have a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathcal{X}$, the instance space, and $y_i \in \mathcal{Y} = \{1, 2, \dots, K\}$, the classes. We assume that the true hypothesis h^* is contained in the hypothesis space $\mathcal{H}$ which is explored by our learner.

In this setting, we can quantify uncertainty of probabilistic, Bayesian, and Levi learners [7]. Probabilistic learners produce only a single hypothesis, so their total uncertainty is completely aleatoric. The uncertainty quantification of probabilistic learners is therefore not affected by biased EU estimates. Levi learners [15] produce a set of possible hypotheses called the *credal set* [16, 22], but do not define a probability distribution on this set. This leads to order statistic-based expressions for epistemic uncertainty [7] which are not commonly used in practice. Our work is therefore focused on Bayesian learners which, given training data $\mathcal{D}$, produce a posterior distribution $p(h|\mathcal{D})$ on the set of possible hypotheses.

Given an instance x , a hypothesis h_i produces a probability distribution $h_i(x) = \theta^{(i)} = (\theta_1^{(i)}, \theta_2^{(i)}, \dots, \theta_K^{(i)}) \in \Theta$ over the classes. At test time, we make use of the posterior produced by our Bayesian learner, $p(h|\mathcal{D})$, to obtain a distribution over Θ :

$$Q(\theta) = \int_{\mathcal{H}} \delta(h(x) - \theta) dp(h|\mathcal{D})$$

where $\delta(\cdot)$ denotes the Dirac delta function. To make a prediction, we ideally use the averaging estimator $\bar{\theta} = \int h(x) dp(h|\mathcal{D})$, but in practice, often use a finite ensemble $\{h_1, h_2, \dots, h_M\}$ of models to approximate the integral by a sample mean: $\bar{\theta} = \frac{1}{M} \sum_{i=1}^M h_i(x)$.

We now turn to scoring rules [19, 4] and their application to uncertainty quantification. A scoring rule is a function $\ell : \Theta \times \mathcal{Y} \rightarrow \mathbb{R}$ such that

$$L_\ell(\bar{\theta}, \theta) := \mathbb{E}_{y \sim \theta}[\ell(\bar{\theta}, y)] \quad (1)$$

is well-defined for all $\bar{\theta}, \theta \in \Theta$. A scoring rule which satisfies $L_\ell(\theta, \theta) \geq L_\ell(\bar{\theta}, \theta)$ for all $\bar{\theta}, \theta \in \Theta$ is called proper, and strictly proper if the inequality is tight if and only if $\theta = \bar{\theta}$ [4]. If we choose ℓ to be a loss function, such as the log-loss $\ell(\bar{\theta}, y) = -\log(\bar{\theta}_y)$, then (1) induces a natural measure of total uncertainty:

$$TU(Q) = \mathbb{E}_{\theta \sim Q}[L_\ell(\bar{\theta}, \theta)]. \quad (2)$$

In words, scoring rule-based total uncertainty measures the expected loss incurred by the use of the averaging estimator. If we subtract from (2) the expected loss incurred by use of the true hypothesis, $AU(Q) = \mathbb{E}_{\theta \sim Q}[L_\ell(\theta, \theta)]$, we are left with a measure of epistemic uncertainty. In the case of the log-loss, this is

$$EU(Q) = TU(Q) - AU(Q) = \mathbb{E}_Q[KL(\theta || \bar{\theta})], \quad (3)$$

which is equivalent to the classical entropy-based uncertainty decomposition [2]. We use the log-loss as a running example, and later show how to apply our reasoning to another scoring rule.

We note that it has been proposed to take an additional expectation over Q , where the subject of the expectation replaces $\bar{\theta}$ in the formula for $EU(Q)$ by [7]. However, the same authors have since published work in which they focus on the averaging estimator exclusively [8]. The averaging estimator also coincides better with the more general literature, in spite of the criticisms made by [23].

At this time, we have not clarified whether $\bar{\theta}$ in the previous definitions refers to the true model average $\int h(x) dp(h|\mathcal{D})$, or the prediction made by a realized ensemble, namely $\frac{1}{M} \sum_{i=1}^M h_i(x)$. Clarifying this is necessary to define our target for estimation. While the definitions coincide asymptotically, low values for M (e.g. 5-10) are standard in uncertainty quantification, and the systematic discrepancy is significant in that regime (see figure 1). Here, we target the finite ensemble version of $\bar{\theta}$ because it is consistent with the notion of the expected loss of our predictions, and this notion is relied upon in most

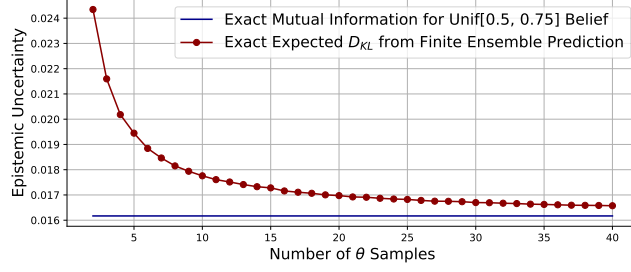


Figure 1: Comparison of the two alternative definitions of log-loss epistemic uncertainty for a uniform belief over $[0.5, 0.75]$ of the Bernoulli parameter, averaged over 10^4 finite ensembles.

uncertainty quantification-related downstream tasks. Hence, we attempt to estimate

$$\mathbb{E}_{\theta \sim Q} \left[KL(\theta || \frac{1}{M} \sum_{i=1}^M h_i(x)) \right] \quad (4)$$

for the log-loss epistemic uncertainty, and

$$\mathbb{E}_{\theta \sim Q} \left[\sum_{k=1}^K (\theta_k - \frac{1}{M} \sum_{i=1}^M h_i(x)_k)^2 \right] \quad (5)$$

for the Brier score epistemic uncertainty.

3 Corrected Estimators

In this section we compile corrected estimators for the log-loss and Brier scoring rules. We begin by analyzing the resubstitution bias for log-loss epistemic uncertainty. The resubstitution estimator for (4) is

$$\frac{1}{M} \sum_{i=1}^M KL(\theta^{(i)} || \frac{1}{M} \sum_{j=1}^M \theta^{(j)}).$$

While this estimator is consistent by the law of large numbers [1], it will clearly underestimate its target. Using a first-order Taylor approximation of the logarithm around $ME[\theta_k]$, we approximate its bias as follows:

$$\mathbb{E} \left[\sum_{k=1}^K \left((\theta_k - \theta_k^{(1)}) \log \left(\sum_{j=1}^M \theta_k^{(j)} \right) \right) \right] \approx - \sum_{k=1}^K \frac{Var_Q[\theta_k]}{ME[\theta_k]}.$$

Using sample mean and sample variance $\hat{\sigma}_k^2$, we obtain the corrected estimator

$$\widehat{EU}_{LogLoss} = \frac{1}{M} \sum_{i=1}^M KL(\theta^{(i)} || \frac{1}{M} \sum_{j=1}^M \theta^{(j)}) + \alpha \sum_{k=1}^K \frac{\hat{\sigma}_k^2}{\sum_{i=1}^M \theta_k^{(i)}}, \quad (6)$$

where α controls the bias-variance tradeoff. For the Brier score, it is straightforward to verify that the resubstitution bias is given by $-\frac{2}{M} \sum_{k=1}^K \text{Var}_Q[\theta_k]$ exactly. Plugging in the sample variance yields

$$\widehat{EU}_{Brier} = \left(\frac{1}{M} + \frac{2\alpha}{M^2 - M} \right) \sum_{i=1}^M \sum_{k=1}^K (\theta_k^{(i)} - \frac{1}{M} \sum_{j=1}^M \theta_k^{(j)})^2 \quad (7)$$

as the corrected estimator of (5). Note that optimizing the mean squared error with respect to α will generally require us to compute fourth order moments. Due to our focus on small sample sizes, we do not attempt to do so, but instead use $\alpha = \frac{1}{2}$ as an uninformed choice.

4 Numerical Examples

4.1 Analytically Tractable Cases

We begin by modeling a Bernoulli random variable to judge estimation methods on the following sources of epistemic uncertainty:

1. **Ignorance**, which we model by a $\mathcal{U}[0.2, 0.7]$ belief over the Bernoulli parameter θ .
2. **Strong disagreement between hypotheses**, which we model by a $\frac{1}{2}\delta((1 - 10^{-6}) - \theta) + \frac{1}{2}\delta(10^{-6} - \theta)$ belief.
3. **Weak disagreement between hypotheses**, which we model by a $\frac{1}{2}\delta(0.56 - \theta) + \frac{1}{2}\delta(0.55 - \theta)$ belief.

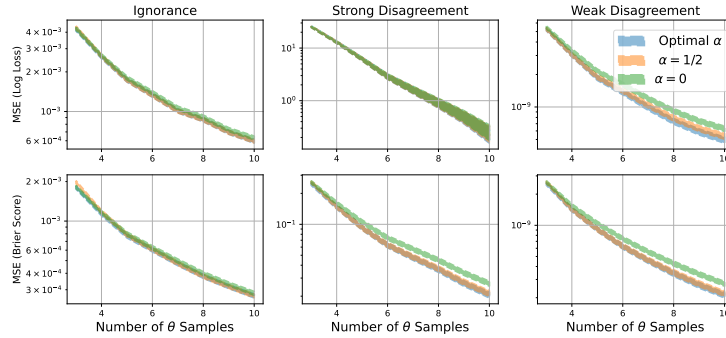


Figure 2: Two-sided 95% t-confidence intervals for the mean squared estimation error.

Figure 3 compares the estimation error achieved by the resubstitution estimator with the corrected estimator. We consider both the uninformed choice $\alpha = 1/2$ and the MSE-minimizing choice for α . We observe a significant reduction in

MSE due to the corrected estimator in 3 cases: weak disagreement with log-loss scoring, and strong or weak disagreement with Brier scoring. The reduction achieved by $\alpha = 1/2$ is comparable to the reduction achieved by the optimal α in all cases.

4.2 Deep Ensembles

In this section, we study the estimation of total uncertainty (see supplementary material for details on total uncertainty estimation) associated with predictions made by an ensemble of ResNet-8s [5]. We train 20 ResNets on the CIFAR-10 dataset [12], up to 10 of which are intended for prediction, and an additional 10 allow us to construct an unbiased Monte Carlo estimator (with smaller variance) of the estimation target. We use the Monte Carlo estimator as a proxy for the estimation target to approximate mean squared estimation errors, and as an oracle for downstream tasks.

Estimation Error. We begin by inspecting the improvement in mean squared error on CIFAR-10 (In-Distribution data), as well as on the Out-of-Distribution datasets CIFAR-100 and SVHN [18]. We use $\alpha = 1/2$ as an uninformed choice and an oracle value of α determined by a grid search for the minimizer of the MSE to determine the benefit of corrected estimators. The

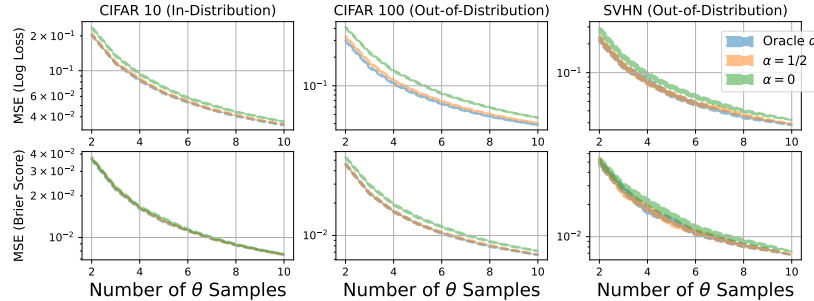


Figure 3: 95% t-confidence intervals for the mean squared error using the empirical distribution (without replacement) of θ (200 runs).

results are shown in figure 3. Improvement in MSE due to bias corrections is largest on Out-of-Distribution data, where the MSE achieved by $\alpha = 1/2$ is reduced by (21% (log-loss), 15% (Brier score)) for CIFAR-100 ($M = (3, 3)$) and (19%, 16%) for SVHN ($M = (2, 5)$). On In-Distribution data, we observe that corrected estimators improve upon resubstitution estimates by 13% on log-loss uncertainty ($M = 5$) and do not significantly improve on Brier score uncertainty. Furthermore, similarly to the analytically tractable cases, we find that $\alpha = 1/2$ yields comparable improvements to the oracle value of α in most cases.

Out-of-Distribution detection. We now consider downstream task performance on Out-of-Distribution (OoD) detection [6]. OoD AUC scores [8] achieved by the use of uncorrected, corrected, and Monte Carlo estimators are shown

in figure 4. There is no clear improvement in the average AUC achieved by $\alpha = 1/2$, in spite of the improvement in MSE shown earlier. Use of the oracle Monte Carlo estimator however does lead to better OoD performance.

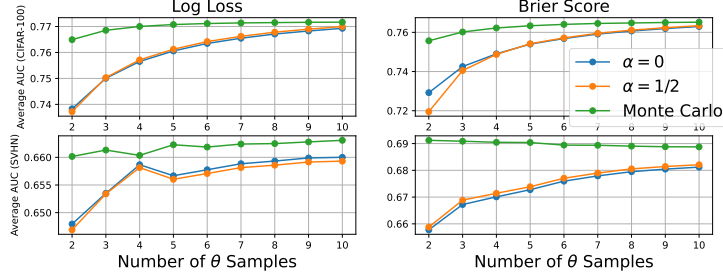


Figure 4: Average AUC on Out-of-Distribution Detection using the empirical distribution (without replacement) of θ (500 runs).

Selective prediction. We judge selective prediction performance based on the area under the loss-rejection curve (AULC) [8] on CIFAR-10 classification. Average AULC values are shown in figure 5. For selective prediction, the Monte

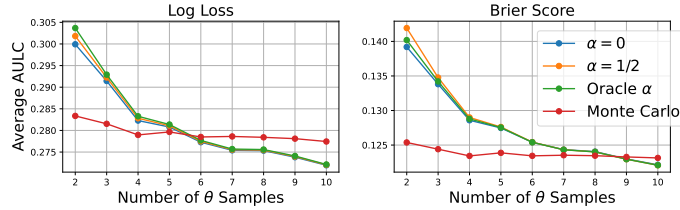


Figure 5: Average AULC on Selective Prediction using the empirical distribution (without replacement) of θ (50 runs).

Carlo estimator leads to better performance only for moderate sample sizes. The $\alpha = 1/2$ and oracle α estimators do not yield any improvements.

5 Discussion

We have presented corrected estimators for log-loss and Brier score epistemic uncertainty. Through a simple, uninformed choice of the bias-variance tradeoff, these estimators were able to reduce the mean squared estimation error by up 13% on In-Distribution data and 21% on Out-of-Distribution data. Furthermore, we found that the reduction in estimation error depends on the source of epistemic uncertainty, as well as the scoring rule used.

However, we did not find a significant improvement in Out-of-Distribution detection, even though the estimation target proxy on which the MSE was

significantly reduced did significantly improve upon the uncorrected estimator. We hypothesize that this is due to the fact that most downstream tasks, including the ones considered here, depend on accurate estimation of the difference in uncertainty between different instances: increasing all estimates by a positive correction only helps if these increases introduce sufficient further differentiation between instances in comparison to the additionally introduced variance.

We also did not find a significant improvement in selective prediction. We hypothesize that this is due to similar reasons as outlined for OoD. Furthermore, we found that the Monte Carlo estimator is outperformed by (corrected) resubstitution estimators at large sample sizes. We hypothesize that this is due to the fact that log-loss and Brier score incorporate the amount disagreement in the predicted class probability distribution, and this disagreement naturally correlates more strongly with the resubstitution estimate than with the Monte Carlo estimate. The behavior of the AULC when loss is measured using the zero-one loss instead supports this hypothesis (see supplementary material).

6 Limitations and Future Work

Our study suffers from the following limitations:

- **Missing scoring rules.** We did not derive bias corrections for several other scoring rules that could be used for uncertainty quantification, such as zero-one or spherical loss functions.
- **Small model.** ResNet-8 is the smallest version of the ResNet architecture and performs significantly worse on CIFAR-10 classification than deeper counterparts. It is possible that some of our observations do not generalize to significantly larger architectures or other classification tasks.
- **Single Ensemble.** We performed only a single ensemble training run and then sampled from the empirical distribution of models produced by this run. Performing several runs would reduce the variance in our experiments.

This research is a work-in-progress and future work will address the above limitations and aim for a more clear and comprehensive overview of corrected epistemic uncertainty estimation. Furthermore, future work will include methodology on the estimation of differences in uncertainty to improve downstream task performance.

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A ResNet8 Training

We trained 20 ResNet8s with different random initializations by use of the following PyTorch architecture:

```
class ResNet8(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = torch.nn.Conv2d(in_channels = 3, out_channels = 16,
            kernel_size = (3, 3), stride = (1, 1), padding = (1,1), bias = False)

        self.bn1 = torch.nn.BatchNorm2d(num_features = 16)

        self.relu = torch.nn.ReLU(inplace = True)

        self.bb1 = torchvision.models.resnet.BasicBlock(16, 16)

        self.bb2 = torchvision.models.resnet.BasicBlock(16, 32, stride = 2,
            downsample = torch.nn.Sequential(torch.nn.Conv2d(16, 32, stride = (2,2),
            kernel_size = (1,1), bias = False), torch.nn.BatchNorm2d(32)))

        self.bb3 = torchvision.models.resnet.BasicBlock(32, 64, stride = 2,
            downsample = torch.nn.Sequential(torch.nn.Conv2d(32, 64, stride = (2,2),
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kernel_size = (1,1), bias = False), torch.nn.BatchNorm2d(64)))

self.avgpool = torch.nn.AdaptiveAvgPool2d(output_size = (1,1))
self.linear = torch.nn.Linear(64, 10, bias = True)

def forward(self, x):
    out1 = self.relu(self.bn1(self.conv1(x)))
    outbb = self.bb3(self.bb2(self.bb1(out1)))
    return self.linear(self.avgpool(outbb).squeeze())

```

We ran training for $2 \cdot 10^4$ epochs with a learning rate of 0.1. We used minibatch (batch size 128) gradient descent with a momentum of 0.9 and a weight decay of 10^{-4} to optimize cross-entropy loss.

B Total Uncertainty Estimation

Table 1 compiles log-loss and Brier score estimation targets as well as the corresponding (corrected) resubstitution and Monte Carlo estimators used in this paper.

Table 1: Total Uncertainty Targets and Estimators. $S(\theta)$ denotes Shannon entropy [21], $KL(\theta^{(1)}||\theta^{(2)})$ denotes the Kullback-Leibler divergence [13] of $\theta^{(1)}$ from $\theta^{(2)}$, and $\bar{\theta}$ denotes $\frac{1}{M} \sum_{i=1}^M \theta^{(i)}$.

Log Loss	
Target	$\mathbb{E}_{\theta \sim Q}[S(\theta) + KL(\theta \bar{\theta})]$
Corrected	$\frac{1}{M} \sum_{i=1}^M S(\theta^{(i)}) + KL(\theta^{(i)} \bar{\theta}) + \alpha \sum_{k=1}^K \frac{\hat{\sigma}_k^2}{M\bar{\theta}_k}$
Monte Carlo	$\frac{1}{M} \sum_{i=M+1}^{2M} S(\theta^{(i)}) + KL(\theta^{(i)} \bar{\theta})$
Brier Score	
Target	$\mathbb{E}_{\theta \sim Q}[1 - \sum_{k=1}^K \theta_k^2 + (\theta_k - \bar{\theta}_k)^2]$
Corrected	$1 - \sum_{i=1}^M \sum_{k=1}^K (\theta_k^{(i)})^2 + (\frac{1}{M} + \frac{2\alpha}{M^2-M})(\theta_k^{(i)} - \bar{\theta}_k)^2$
Monte Carlo	$1 - \frac{1}{M} \sum_{i=M+1}^{2M} \sum_{k=1}^K (\theta_k^{(i)})^2 + (\theta_k^{(i)} - \bar{\theta}_k)^2$

C Zero-One Loss for Selective Prediction

Areas under the zero-one loss curve are shown in figure 6. The Monte Carlo estimator maintains the lowest AULC even at larger sample sizes.

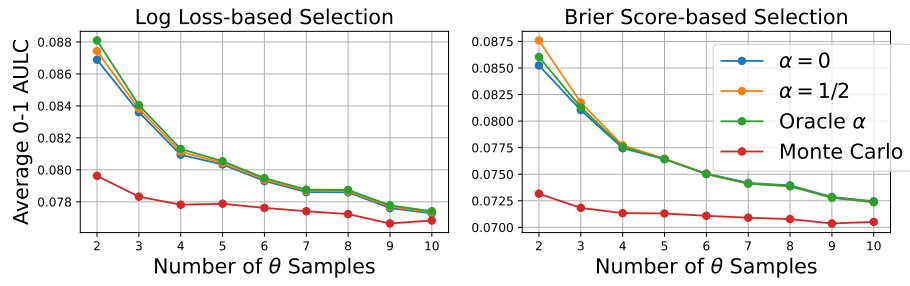


Figure 6: Average areas under the zero-one loss curve using the empirical distribution (without replacement) of θ (50 runs).